

Alan Bao

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EDUCATION

Northeastern University, Boston MA

May 2027

College of Engineering

GPA: 3.6 (Dean's List: Spring 2024, Fall 2024, Spring 2025)

Candidate for Bachelor of Science in Computer Engineering and Computer Science

Relevant Courses: Fundamentals of Electronics, Computer Architecture and Organization, Circuits and Signals, Fundamentals of Digital Design, Intro to Program Design and Implementation, Embedded Design: Enabling Robotics, Computing Fundamentals for Engineers, Fundamentals of Networks, Discrete Structures

Activities: Undergraduate Research, Northeastern Robotics, Engineers without Borders, NU Wind Ensemble, NU Orchestra

TECHNICAL KNOWLEDGE

Coding: C#, C++, Python, SystemVerilog, VB6, Linux

Electronics: SparkFun, wiring, basic soldering, PCB design and testing

Software: Visual Studio 2022, SolidWorks, AutoCAD, MATLAB, Altium, Vivado, PSpice

WORK EXPERIENCE

Software Engineer Co-op | Veeco Instruments, Horsham PA

July - December 2025

- Developed C# application for validating and testing product configuration files
- Wrote tool communication contracts and implemented them in WCF and gRPC APIs
- Upgraded configuration files and set up tool PCs for customers
- **Automated Testing Project**
 - Designed an object oriented application that automatically tests tool configuration file quality by remotely controlling a tool control software
 - Developed gRPC and WCF Client-Host systems to expose an API
- **Configuration Utility Project**
 - Refactored existing utility application to implement a more scalable design
 - Added validation rules for tool configuration files by text parsing through the files

PROJECTS

Analog Computing and Energy Efficient Circuits | Northeastern Undergraduate Research

April 2025 - Present

- Exploring medical and machine learning applications of analog design for ultralow power consumption
- Analyzing machine learning models with PyTorch

Mitosis: 3D Printed 3D Printer | Northeastern Robotics

Jan 2025 - Present

- **Objective** - Design a 3D printer out of only 3D printed parts
- Tested and developed C++ code for the stepper motor and driver
- Designed PCB to access multiple motors with a motor driver using Altium Designer

Spider Robot | Embedded Design: Enabling Robotics

Fall 2024

- Programmed with programmable logic (FPGA) and C++ to control 18 servo motors and implement programmable logic to generate PWM signals

Wastewater Management System | Engineers without Borders

Jan - Nov 2025

- Wrote reports for post-implementation assessments